Graph-Grounded Layered Instruction Fine-Tuning of Small Language Models for Cyber Threat Intelligence

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***Abstract*—Vulnerability response in security operations is a chain of mapping decisions: a newly disclosed CVE must be scored for severity and exploitability, traced to its weakness class and the adversary techniques it enables, and linked to mitigations and detections before a patch or compensating control is scheduled. Because the asset and exposure context behind these decisions often cannot leave the organization, locally served small language models (SLMs)—here, open-weight models of 8B to 32B parameters that a two-GPU host can fine-tune and serve—are attractive, but only if their outputs land on the correct catalog identifier and their training targets are traceable to source intelligence. We present an SLM specialization workflow built on a versioned Neo4j graph joining twelve public threat and vulnerability intelligence sources (e.g., NVD, CISA KEV, EPSS, CWE, ATT&CK) (*Ariadne*), and a template engine that materializes each instruction target by executing a typed graph query (*Sophia*). Using a functionally optimized approach, we use a layered Instruction Fine-Tuning (IFT) template-based strategy to train Qwen2.5-32B-Instruct by which we increase attack-technique extraction scores from 18.2 to 65.8, vulnerability mitigation selection scores from 7.6 to 62.5, and the three-benchmark mean we use for testing from 57.7 to 65.0 (95% bootstrap interval 64.1–66.0), with an 11.6-point decrease on the MMLU-Pro general-knowledge benchmark. Against the single-stage model that the first layer produces, the three later layers add 3.2 points to the three-benchmark mean but cost 2.3 points on the task-aligned benchmark.**

***Index Terms*—small language models, SLMs, cyber threat intelligence, CTI, vulnerability management, cyber security, instruction fine-tuning, IFT, knowledge graphs, data provenance**

I. Introduction

For each newly disclosed vulnerability (a Common Vulnerabilities and Exposures entry, CVE [1]) found in a scan of a given environment, security operations center (SOC) analysts must join cyber threat intelligence (CTI) from separate, largely public catalogs: the CVSS severity vector from the National Vulnerability Database (NVD) [2], exploitation likelihood from the Exploit Prediction Scoring System (EPSS) [3], confirmed exploitation from the CISA Known Exploited Vulnerabilities (KEV) catalog [4], public exploit code [5], [6], the weakness class (CWE [7]) and attack patterns (CAPEC [8]) behind the flaw, the ATT&CK techniques it enables [9], and the related mitigations, D3FEND countermeasures [10], and Sigma detections [11]. This analysis must then be combined with sensitive local context (asset inventory, scanner findings, exposure, business criticality) to plan the optimal risk response (patch now, apply a compensating control, or add detection until a maintenance window), a per-vulnerability analysis that NIST notes organizations lack the time and expertise to perform manually [12].

Each join maps free text into a controlled vocabulary, and near-misses are costly: a parent ATT&CK technique in place of the sub-technique points detection engineering at the wrong telemetry; a related but incorrect CWE suggests the wrong compensating control; a wrong CVSS Attack Vector moves a vulnerability between remediation tiers; a misnormalized threat actor misdirects hunting; and an unparseable answer breaks downstream ticketing and orchestration. CTIBench [13] and AthenaBench [14] test models on exactly these mappings: CTI knowledge questions (CKT), attack-technique extraction from a threat description (ATE), root-cause mapping of a CVE to its CWE (RCM), risk-mitigation selection for a technique (RMS), vulnerability-severity prediction as a CVSS vector (VSP), and threat-actor attribution (TAA) (Section V).

That private context, which may include regulated data such as PHI and PII, along with licensed intelligence under sharing restrictions, is exactly what contractual, confidentiality, and data-residency obligations keep away from hosted large language model (LLM) APIs, frontier models included, favoring locally served open-weight models. Throughout, *small language model* (SLM) is an operational term, not a parameter threshold: an open-weight model that an organization can fine-tune and serve on its own hardware, here on two 80-GB GPUs. The primary base model has 32B parameters, and we also test 8B, 14B, and sparse 30B (3B active) bases.

The standard way to specialize such a model is *instruction fine-tuning* (IFT): continued supervised training on (instruction, input, output) *triples*, so that the model learns to produce the output when given the instruction and input. Applying IFT to CTI raises two problems. First, the outputs must be correct and traceable: a recommendation that justifies an emergency

change or a patch exception must lead back to a source record, yet triples synthesized by another language model are hard to trace to any source fact. Second, a model fine-tuned for several analyst tasks can lose one capability while gaining another, and a single final score hides such losses.

We address the first problem with two components. *Ariadne* is a versioned Neo4j knowledge graph that merges the twelve public sources above into one typed schema. *Sophia* generates IFT triples from expert-written templates: each template compiles to a graph query, and its placeholders are filled from the query's results, so every training target is derived from specific graph records rather than written by a language model. Private entities such as assets and scanner findings fit the same schema but are not used here. We address the second problem with *layered IFT*: rather than one pass over a mixed corpus, the model is fine-tuned in four successive layers (catalog mapping, threat-actor attribution, SOC-style structured response, and recalibration), each starting from the previous layer's checkpoint, and every intermediate checkpoint is retained and evaluated.

Our contributions are: (i) a graph-grounded data pipeline that produced a 219,675-row core training corpus in which every row resolves to its template, graph binding, and public source identifiers; (ii) a layered IFT recipe that, applied to Qwen2.5-32B-Instruct, raises attack-technique extraction from 18.2 to 65.8 and mitigation selection from 7.6 to 62.5, lifting the three-benchmark mean from 57.7 to 65.0 (95% interval 64.1–66.0), within 1.0 point of DeepSeek-V3.2-Exp, at a cost of 11.6 points on MMLU-Pro; against the single-stage first layer, the three later layers add 3.2 points to that mean but cost 2.3 on AthenaBench; (iii) a checkpoint-level diagnosis of interference: the attribution layer raises canonical threat-actor attribution by 18.4 points, while the SOC-oriented layer raises CyberSOCEval malware accuracy by 19.7 points but erases 33.5 points of canonical attribution; and (iv) an overlap audit at the 13-gram, entity, and complete-answer levels, extended after training to CyberSOCEval and the general instruction data, that finds little verbatim overlap and does not establish leakage as the explanation for the gains. The pipeline is released as Glaukopis [15].

II. Related Work

A. Instruction Fine-Tuning and Sequential Specialization

IFT adapts a pretrained language model by continuing training on examples that pair a task instruction, and optionally an input, with the desired response; the loss is computed on the response tokens, so the model learns to produce the response given the instruction. FLAN showed that IFT on many instruction-formulated tasks transfers to unseen tasks [16]; InstructGPT combined supervised demonstrations with preference-based training [17]; and later work studied how IFT scales with the number and mixture of tasks [18]. Continued training on domain text further improves domain tasks, motivating a separation between broad domain absorption and narrow task behavior [19]. Fine-tuning one model sequentially on several datasets, however, exposes it to *catastrophic interference*: learning a later task can overwrite behavior acquired earlier [20], and in a controlled comparison of mixed and sequential supervised fine-tuning, sequential training forgot earlier skills [21]; a layered schedule must therefore be judged against a single-stage alternative. Layered IFT is a sequential schedule designed to make this interference observable rather than to eliminate it.

B. Security Instruction Tuning

CyberPal.AI's SecKnowledge pipeline combines expert-defined schemas, structure-driven generation, BRON graph paths, and LLM-augmented content generation [22]; follow-on work develops cybersecurity-expert SLMs with an enriched instruction pipeline using expert steering and LLM-driven multi-step grounding [23]. SEvenLLM builds a bilingual CTI instruction corpus and benchmark [24], and CyberBench pairs a multi-task cybersecurity benchmark with instruction-tuned CyberInstruct models [25]. These systems motivate domain-specific supervision, but none makes every training label the deterministic result of an executed graph query (Table I).

C. CTI Knowledge Graphs

Open-CyKG constructs an open cyber-threat knowledge graph from unstructured reports [26]. That work concerns how facts enter a graph; this work concerns how a populated graph is turned into IFT triples. The two qualities should not be conflated: a Sophia target is traceable to its graph binding, but the graph can still contain incomplete, stale, or misinterpreted source facts.

III. Graph-Grounded Training Data

A. Formulation

An IFT triple $r = (i, x, y)$ consists of an instruction i, an optional input or question x, and a target output y. Let $G_t = (V_t, E_t, A_t)$ be the Ariadne graph at snapshot t, with typed nodes V_t, typed relationships E_t, and node attributes A_t. A Sophia template τ is text with typed placeholders in its instruction, question, and answer parts (i_τ, x_τ, y_τ); it compiles to a graph query q_τ that includes any constraints the template declares. Executing the query returns a set of bindings $q_\tau(G_t)$, each binding b assigning a node, relationship, or attribute value to every placeholder. For each binding $b \in q_\tau(G_t)$, Sophia emits one triple

$$r_{\tau,b} = \big(\operatorname{render}(i_\tau, b),\ \operatorname{render}(x_\tau, b),\ \operatorname{render}(y_\tau, b)\big), \quad (1)$$

where $\operatorname{render}(s, b)$ replaces each placeholder in text s with its value under b. In short, a template generates one IFT triple per query result. Each row also records its template identifier and source tags, linking it to the template, snapshot, and public identifiers it was built from.

The fine-tuned model $p_\theta(y \mid i, x)$ is trained with the standard next-token cross-entropy on the target tokens. For layer s with dataset D_s,

$$\mathcal{L}_s(\theta) = -\sum_{(i,x,y)\in D_s} \sum_{j=1}^{|y|} \log p_\theta(y_j \mid i, x, y_{<j}), \quad (2)$$

TABLE I

POSITIONING AGAINST CLOSELY RELATED SECURITY INSTRUCTION-TUNING SYSTEMS. "PARTIAL" INDICATES STRUCTURED SCHEMAS, GRAPH PATHS, OR GROUNDING TRACES FOR PORTIONS OF THE PIPELINE WITHOUT MAKING EVERY GENERATED LABEL THE DETERMINISTIC RESULT OF AN EXECUTED TYPED GRAPH QUERY.

System	Graph-grounded targets	LLM-generated training text	Retained layer checkpoints	Provenance of a training label
CyberPal.AI / SecKnowledge [22]	Partial: expert schemas and BRON graph paths for subsets	Yes: content-grounded LLM augmentation	Multi-phase generation; stage checkpoints not analyzed	Source/schema traces; some explanations LLM-generated
Cybersecurity-expert SLMs [23]	Partial: LLM-driven multi-step grounding	Yes: enriched reasoning traces	Enrichment pipeline; layer effects not isolated	Enriched traces with expert-in-the-loop steering
This work	Yes: typed Ariadne queries materialize every Sophia CTI row	No for Sophia targets; auxiliary general mixtures are external	Yes: all four layer checkpoints retained and evaluated	Row resolves to template, graph binding, and public-source keys

TABLE II

ARIADNE LAYERS, SOURCES, PRINCIPAL ENTITY TYPES, AND THE MAIN CROSS-FRAMEWORK LINKS THAT TEMPLATES TRAVERSE.

Layer	Sources	Principal entities	Main links
Adversary behavior	ATT&CK, Engage	techniques, groups, software, mitigations; engagement activities	technique–mitigation; technique–group
Weakness & attack pattern	CWE, CAPEC	weaknesses; attack patterns	CAPEC→CWE; CAPEC→technique
Vulnerability	CVE, NVD	CVE records; CVSS vectors; affected products	CVE→CWE; CVE→CAPEC
Exploitation signal	CISA KEV, EPSS	KEV entries; exploitation probabilities	→CVE
Exploit evidence	ExploitDB, PoC-in-GitHub	exploit and proof-of-concept records	→CVE
Countermeasure	D3FEND	defensive techniques	D3FEND–technique
Detection	Sigma	detection rules	rule→technique

with the checkpoint from layer $s-1$ initializing layer s. No new optimizer, loss, or architecture is introduced; the method lies in how each D_s is generated and in how the layers are sequenced.

B. Ariadne: Graph Schema

Ariadne is a Neo4j property graph built from the twelve public sources in Table II, grouped into seven layers. Each source contributes typed nodes keyed by their natural identifiers (STIX, CVE, CWE, CAPEC, or D3FEND IDs). Ingestion uses Neo4j `MERGE` operations under uniqueness constraints on those keys, so re-running it never duplicates nodes, and it follows dependency order—framework catalogs, then vulnerabilities, then exploit evidence—so every cross-framework relationship has both endpoints present. The relationships most used by templates form a spine from vulnerability to weakness to attack pattern to adversary technique, with mitigations, countermeasures, and detections attached to techniques and exploitation signals attached to vulnerabilities. The snapshot used here holds 214,651 nodes with 31 labels and 330,935 relationships of 41 types. Bounded catalogs are loaded in full; the high-churn vulnerability and exploit feeds are scoped to 2024 onward. EPSS and the current-year NVD feed are refreshed daily and other sources on explicit refresh, so dataset freshness is tracked per source.

C. Sophia: Template-Driven Triple Generation

A Sophia template is plain text with placeholders in braces. A placeholder {a:Label.prop} binds alias a to a node of type Label and inserts its property prop; {a.rel>Label.prop} follows relationship rel from the node bound to a; and {a.prop} reuses an existing binding. Hidden sections <*{...}*> add bindings or constraints to the query without producing text. The language also supports any/all filters over list-valued properties, list expansion, and source prefixes that distinguish same-named types such as ATT&CK and CAPEC attack patterns. A simple template binds one ATT&CK technique and emits its properties:

```
Question: Explain ATT&CK technique
{ap:attack-pattern.name} ({ap.id}) and how it is
used.
Answer: Technique {ap.name} ({ap.id}) is described
as: {ap.description}. Commonly observed across
{ap.x_mitre_platforms}.
```

Executed against Ariadne, this template yields one triple per technique node. Templates become more useful when they traverse relationships across frameworks. The following template from the generator's library correlates a CVE with its attack pattern and weakness:

```
Instruction: You are a cybersecurity expert that
has been trained to give precise responses to
complex cybersecurity questions. You work in a SOC
protecting data for enterprise customers helping
to protect their digital assets.
Question: Correlate CVE, CWE, and CAPEC for a
single vulnerability chain.
Answer: CVE {cve:CVE.id} impacts CAPEC
{cap:cve.impacts>CAPEC.name} and is categorized
as weakness {cwe:cve.problemType>Weakness.name}.
CAPEC {cap.id} exploits weakness
{cwe2:cap.exploits>Weakness.name}. <*{force
cwe.id=cwe2.id}*>
```

The template compiler turns each binding into a `MATCH` pattern of a Cypher query and the hidden `force` section into a `WHERE` condition, schematically:

```
MATCH (cve:CVE)-[:impacts]->(cap:CAPEC),
  (cve)-[:problemType]->(cwe:Weakness),
  (cap)-[:exploits]->(cwe2:Weakness)
WHERE cwe.id = cwe2.id RETURN cve, cap, cwe, cwe2
```

Only chains in which the CVE's own weakness is also the weakness the attack pattern exploits are returned, and each returned row is rendered into one triple whose answer names the CVE, the attack pattern, and the weakness. Any generated answer can therefore be checked by re-running its query. Negative examples use inequality constraints; because CTI graphs are incomplete, such examples mean "not supported by the graph" rather than "false in the world."

Generation runs in three steps: templates are extracted from their authoring documents; each is executed against Ariadne, producing its triples and a report of failed or empty queries; and the results are merged into one dataset. Rows are then balanced across template families, deduplicated, screened for overlap with evaluation data (Section VI), license-gated, and serialized as JSON. Schema tests catch templates that reference absent node types, relationships, or properties, but they do not replace expert review of whether a template's mapping is semantically right: Sophia guarantees a deterministic path from query to target, not the correctness of every upstream fact.

D. Training Corpus

A *shard* is the set of triples generated for one training layer. The build produced four training shards and three validation shards, the latter holding out 50 rows per template family. The core corpus contained 219,675 rows after balancing; after the overlap screen removed 684 rows, 550 were held out for validation, and one training row identical to a validation row was dropped, it was split into Core-A (168,859 rows: knowledge, multiple-choice, mitigation, attribution, and SOC families) and Core-B (49,581 rows: drills for the RMS, ATE, VSP, and RCM tasks of Table IV). The TAA (attribution) and CSE (SOC-style response) shards contain 20,359 and 17,320 rows. Core-A is mixed with general-purpose instruction data to limit loss of general ability: the first 240,000 rows of the Tülu-3 SFT mixture [27] (239,985 usable conversations) and the 999-row Alpaca demonstration set [28]. These auxiliary examples are not generated by Sophia.

IV. Layered Instruction Fine-Tuning

Layered IFT trains the model in successive layers, each a separate supervised run on its own shard that starts from the previous layer's checkpoint. Fig. 2 summarizes the data, functional areas, quantities, and settings of each layer, and what each retained checkpoint gained or lost. Each layer targets one analyst role, and its checkpoint is retained so that both its gains and any damage to earlier skills can be measured. Checkpoints are named after their layer: Core, +TAA, CSE, and Recal-32B, the final 32B checkpoint.

Layer 1 (Core): catalog mapping. Phase 1A trains on Core-A plus the general instruction data to establish broad CTI knowledge. Phase 1B then trains on Core-B at a lower learning rate with longer, unpacked sequences, so that long structured answers such as CVSS vectors and mitigation lists are learned exactly. (*Packing* concatenates short examples into one training sequence for efficiency; it is disabled where answers must not be split.)

Layer 2 (+TAA): attribution. Drills that map threat-report descriptions to ATT&CK groups.

Layer 3 (CSE): SOC-style response. Drills in the multiple-answer, letter-set format used by CyberSOCEval's malware and threat-intelligence tasks.

Layer 4 (Recal): recalibration. A short pass at a low learning rate that interleaves samples from Core-A, Core-B, and TAA to recover skills disturbed by Layers 2–3.

Ordering from broad catalog knowledge to narrow role-specific behavior lets each layer build on the last, and because each layer consumes a known shard, a regression can be localized to the layer that caused it and addressed by retraining only that layer and its successors.

The base model is Qwen2.5-32B-Instruct [29], trained with LlamaFactory [30] and DeepSpeed ZeRO-3 [31] in bfloat16 with an 8-bit AdamW optimizer on eight 80-GB H100 GPUs. Table III lists per-layer settings. Each shard is consumed in one epoch; Table III gives the examples and tokens each pass consumed. In Phase 1A the 240,000-row sample cap applies to each dataset separately: it truncates Tülu-3 to its first 240,000 rows and leaves Core-A and Alpaca whole, so general data supply 87% of the pass's 314.9M tokens. Layer 4 interleaves about 6,000 examples. Its recipe is model-specific: a lower-learning-rate, more balanced setting that recovered VSP on a 14B base drifted VSP the wrong way at 32B, so the 32B recipe in Table III weights Core-B more heavily; both variants were trained from the same 32B Layer-3 parent for comparison. We also apply the recipe to four other base models (Section VII-F); the 14B transfer uses the alternative Layer-4 setting (learning rate 10^{-6}, mixture 0.25/0.40/0.35, 2,400 examples).

V. Evaluation Setup

AthenaBench contains six task families, each corresponding to one analyst step from Section I (Table IV). A correct RCM answer identifies the weakness class and hence the relevant compensating controls; a correct VSP vector sets remediation priority; and correct ATE and RMS answers direct detection and mitigation work. TAA appears in a lenient *Classic* form and a strict *Canonical* form that requires the group's canonical name or ATT&CK G-code. We also evaluate on CyberSOCEval (malware analysis and threat-intelligence reasoning with multiple-answer questions) [32], CyberMetric (cybersecurity multiple-choice questions in 2K and 10K tiers) [33], and MMLU-Pro, a general-knowledge benchmark used here to measure forgetting [34].

CKT, RCM, CyberMetric, and MMLU-Pro use exact matching, and ATE credits the parent technique. VSP converts predicted and gold vectors to CVSS v3.1 base scores [35] and scores their normalized distance; two of 2,000 final-checkpoint vectors are unparseable and excluded. RMS uses set F1, with half credit for mitigations in the target technique's neighborhood. TAA Classic gives full credit to an alias of the correct group and half credit to a related group; TAA Canonical gives full credit to the canonical name or G-code and half credit to another alias. CyberSOCEval reports strict accuracy and Jaccard similarity over answer-letter sets. These scores differ in kind (Table IV marks each as an accuracy, an F1, or a partial-credit score) and are not calibrated probabilities. Checkpoints are served with vLLM [36], and each score is one evaluation of one training run.

To keep Canonical TAA visible, we average the two TAA forms and define two aggregates:

$$T_{bal} = \tfrac{1}{2}(T_{Classic} + T_{Canonical}), \tag{3}$$

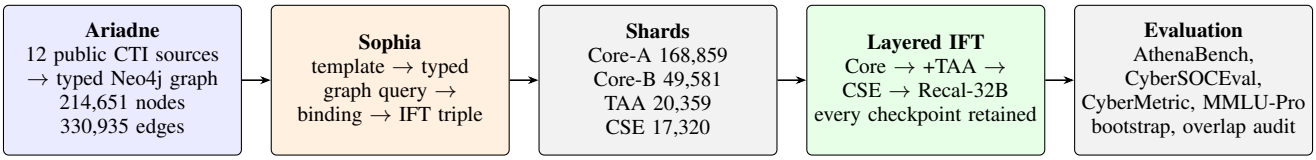


Fig. 1. Data and training pipeline. Ariadne integrates twelve public CTI sources into a versioned typed graph; Sophia executes template queries against it and renders one IFT triple per result; the resulting shards (TAA: threat-actor attribution drills; CSE: SOC-style multiple-answer drills) feed four IFT layers whose checkpoints are all retained and evaluated.

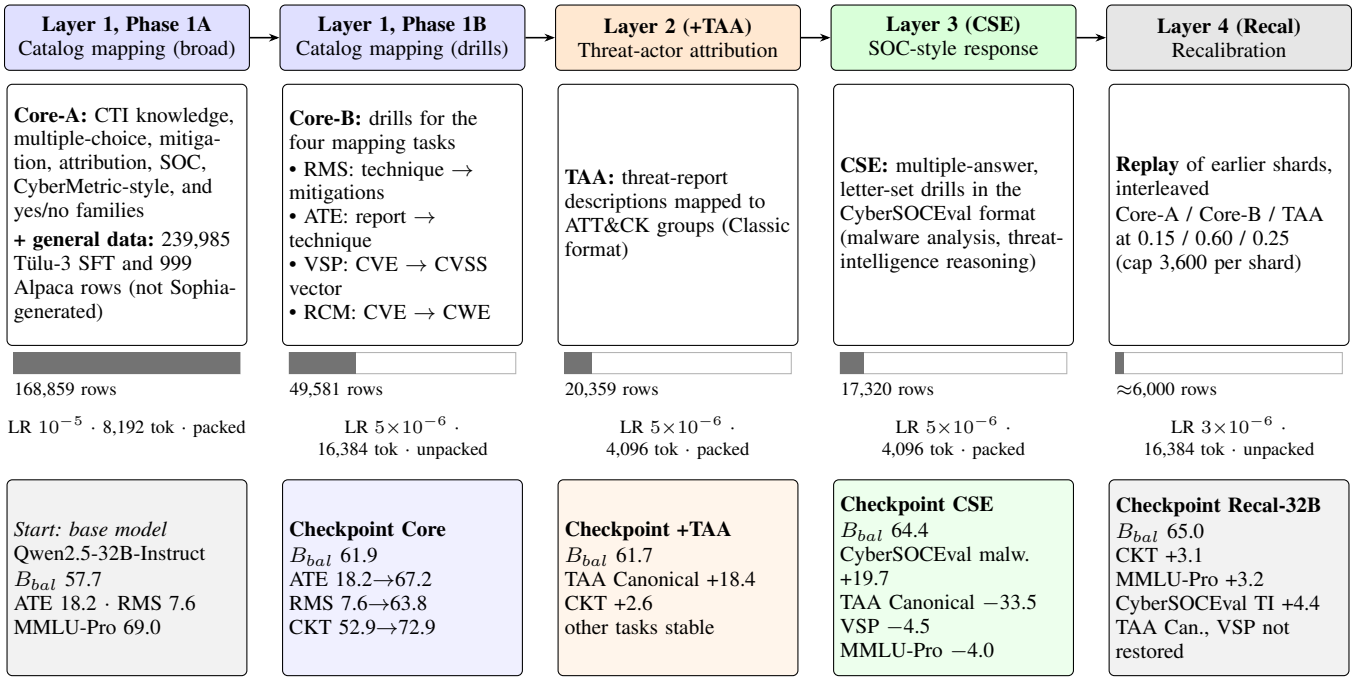


Fig. 2. Layered IFT of Qwen2.5-32B-Instruct. Each layer is a separate supervised run on its own shard (row: training data and functional areas; bar: realized training examples on a common linear scale; then learning rate, maximum sequence length, and packing) that starts from the previous layer's weights. Phases 1A and 1B run back to back and only the merged Core checkpoint is retained. Checkpoint boxes give B_{bal} and the most notable changes; changes for +TAA, CSE, and Recal-32B are relative to the preceding checkpoint (Table VIII).

$$A_{bal} = \tfrac{1}{6}(K + E + R + M + V + T_{bal}), \tag{4}$$

$$B_{bal} = \tfrac{1}{3}(A_{bal} + C + Y), \tag{5}$$

where K, E, R, M, V denote CKT, ATE, RCM, RMS, and VSP; Y is the mean of the CyberMetric tiers; and C is the CyberSOCEval composite (mean of malware strict accuracy, malware Jaccard, and threat-intelligence strict accuracy). T_{bal}, A_{bal}, and B_{bal} are composite averages of these heterogeneous scores, not accuracies; B_{bal} is the three-benchmark mean reported throughout.

For the five fine-tuned checkpoints, 95% percentile bootstrap intervals use 2,000 item-resampling replicates [37]; differences between checkpoints resample the same items for both, giving paired estimates. The intervals capture item-sampling uncertainty for fixed checkpoints, not training-seed variance. Item-level files were unavailable for the base and hosted models, so paired statistics involving them are not reported.

VI. Training–Evaluation Overlap

Because training and evaluation draw on the same public catalogs, we audit overlap at two levels.

Verbatim screen. Before training, every generated row was compared with the locally available evaluation files for shared 13-grams—runs of 13 consecutive lowercased words, the window used in the GPT-3 contamination audit [38]. Rows sharing at least 50 distinct 13-grams with one evaluation item were removed; rows sharing fewer were flagged (Table V). CyberSOCEval files were not present when the CSE shard was built, and the general datasets were not screened, so after training we closed both gaps with the same rule. Against the 1,197 CyberSOCEval questions and their options, no Core or TAA row and 86 of the 17,320 CSE rows share any 13-gram, none more than four. Against AthenaBench, CyberMetric, and CyberSOCEval together, 22 of the 240,000 Tülu-3 rows (none more than twelve) and one of the 999 Alpaca rows do. No row in either screen reaches the removal threshold. This remains a verbatim screen: it cannot detect paraphrase.

TABLE III

CANONICAL QWEN2.5-32B LAYERED IFT SCHEDULE WITH THE EXAMPLES AND TOKENS EACH PASS CONSUMED. TOKENS ARE PROMPT PLUS RESPONSE TOKENS AFTER TRUNCATION TO THE PASS'S MAXIMUM LENGTH, OBTAINED BY REPLAYING LLAMAFACTORY'S ENCODING OF EVERY ROW WITH THE QWEN2.5 TOKENIZER; RESPONSE TOKENS CARRY THE LOSS. LAYER-4 FIGURES ARE EXPECTATIONS OVER ITS INTERLEAVED SAMPLING.

Pass	Layer / input	Examples	Tokens (M)	Resp. (M)	Max tokens	LR	Batch	Packing
1A	Core-A + Tülu-3 SFT + Alpaca	409,843	314.9	175.7	8,192	1×10^{-5}	16	on
	of which Core-A / Tülu-3 / Alpaca	168,859 / 239,985 / 999	40.1 / 274.6 / 0.2					
1B	Core-B	49,581	22.2	7.4	16,384	5×10^{-6}	8	off
2	TAA	20,359	5.0	1.3	4,096	5×10^{-6}	16	on
3	CSE	17,320	7.3	0.3	4,096	5×10^{-6}	16	on
4	Core-A/Core-B/TAA, 0.15/0.60/0.25	≈6,000	≈2.2	≈0.7	16,384	3×10^{-6}	8	off

TABLE IV

EVALUATION TASKS AND THE TYPE OF EACH SCORE. ATHENABENCH TASKS ARE LISTED FIRST.

Task	Model must produce	Items	Score (type)
CKT	answer to a CTI knowledge multiple-choice question	3,000	exact option (accuracy)
ATE	ATT&CK technique described in a threat-report excerpt	500	parent-technique match (accuracy)
RCM	CWE weakness behind a CVE description	2,000	exact CWE (accuracy)
RMS	ATT&CK mitigations for a given technique	500	set F1, half credit for related mitigations (partial credit)
VSP	CVSS v3.1 base vector for a CVE description	2,000	1 − normalized base-score distance (partial credit)
TAA	threat group responsible for a described campaign (Classic / Canonical)	100/133	1 / 0.5 / 0 by group match (partial credit)
CyberSOC-Eval	answer-letter sets for malware and threat-intel questions	609/588	strict accuracy; Jaccard (partial credit)
CyberMetric	multiple-choice answers (2K / 10K tiers)	2,000/ 10,180	exact option (accuracy)
MMLU-Pro	general-knowledge multiple-choice answers	12,032	exact option (accuracy)

Entity and answer exposure. Graph-generated rows can state an evaluated fact in different words, which a verbatim screen misses. We therefore assigned every AthenaBench item to exactly one exposure group: its identifying entity (CVE, technique, or group alias) never appears in training; it appears, but never together with the complete gold answer (*entity only*); or entity and complete gold answer co-occur in one training row (*complete answer*). Table VI reports the final checkpoint's score in each group.

Exposure affects the tasks differently. RCM items whose exact CVE-to-CWE pair appears in training score 66.4, close to the 64.3 of unexposed items and below the 74.2 of items whose CVE appears without the pair. VSP items whose CVE appears score like unexposed items (77.3 vs. 77.9); the seven items whose complete gold vector appears score 89.2, but seven items cannot support a conclusion either way. RMS items whose full mitigation set appears score lower, because those techniques have larger mitigation sets, on which set F1 is harder. ATE accuracy rises with the training frequency of the gold technique, and at the +TAA checkpoint Canonical-TAA items whose alias-to-group mapping appears in training score 52.6 versus 30.6 for the rest. The audit therefore does not establish leakage as the explanation for the gains, but it cannot exclude it: paraphrased and repeated public facts remain uncontrolled.

VII. RESULTS

A. Canonical 32B Chain

Table VII reports the base model and the retained checkpoints. The final checkpoint raises ATE from 18.2 to 65.8, RMS from 7.6 to 62.5, CKT from 52.9 to 75.5, and RCM from 59.5 to 67.1, while VSP decreases by 1.0 point and MMLU-Pro by 11.6. A_{bal} rises from 40.3 to 62.6 (interval 61.5–63.7) and B_{bal} from 57.7 to 65.0 (64.1–66.0).

The improvement is not uniform: AthenaBench contributes about +7.4 points of the 7.3-point change in B_{bal}, while CyberSOCEval stays at 42.6 and CyberMetric moves from 90.2 to 89.8. The gain therefore comes from the benchmark the training shards target most directly, which is evidence of schema adaptation rather than uniform transfer.

B. Layer Transitions

The retained checkpoints show where capabilities move (Table VIII). The TAA layer raises Canonical attribution from 18.4 to 36.8, a paired change of +18.4 points [12.4, 24.4], and CKT by +2.6 [1.0, 4.2], with small changes elsewhere. The CSE layer produces the clearest interference: relative to +TAA, CyberSOCEval malware strict accuracy rises by 19.7 [15.3, 24.2] and threat-intelligence accuracy by 7.9 [3.3, 12.4], but Canonical TAA falls by 33.5 [−39.8, −27.1], VSP by 4.5, CKT by 3.0, and MMLU-Pro by 4.0. Recalibration recovers CKT, threat-intelligence accuracy, and MMLU-Pro, but not Canonical TAA, and leaves VSP below both its parent and the base.

C. Layered versus Single-Stage Training

The Core checkpoint is the model that single-stage fine-tuning on the core shards produces: Layer 1 trains the base once, without the attribution, SOC-format, or recalibration layers, and Core is evaluated with the same harness and per-item records as every later checkpoint. Against it, with paired intervals (Table VIII, last column), the three later layers add 3.2 points to B_{bal} [2.2, 4.2] through CyberSOCEval malware (+18.6) and threat-intelligence (+7.9) accuracy and CKT (+2.7), and remove 2.3 points from A_{bal}, 15.0 from Canonical TAA, 5.3 from VSP, 2.7 from RCM, and 1.8

TABLE V

VERBATIM 13-GRAM SCREEN. "FLAGGED" ROWS SHARE AT LEAST ONE 13-GRAM WITH AN INDEXED EVALUATION RECORD; ROWS AT 50 OR MORE SHARED 13-GRAMS WERE REMOVED.

Shard	Scanned	1	2–4	5–12	13–24	25–49	≥50 removed	Flagged total	Retained
Core	219,675	6,641	10,952	6,656	3,165	6,799	684	34,897 (15.9%)	218,991
TAA	20,509	0	0	0	0	0	0	0	20,509
CSE	17,670	24	17	0	0	0	0	41 (0.2%)	17,670

TABLE VI

TRAINING EXPOSURE OF ATHENABENCH ITEMS AND THE FINAL CHECKPOINT'S MEAN ITEM SCORE IN EACH EXPOSURE GROUP, AS SCORE (n). GROUPS ARE EXCLUSIVE AND COVER ALL ITEMS; THE TWO VSP ITEMS WITH UNPARSEABLE FINAL-CHECKPOINT VECTORS ARE EXCLUDED FROM SCORING, ONE IN EACH OF THE FIRST TWO GROUPS. RMS TECHNIQUES ALWAYS APPEAR IN TRAINING, SO NO ITEM IS UNEXPOSED.

Task	Items	No exposure	Entity only	Complete answer
RCM	2,000	64.3 (1,340)	74.2 (523)	66.4 (137)
VSP	2,000	77.9 (1,553 of 1,554)	77.3 (438 of 439)	89.2 (7)
RMS	500	—	73.1 (227)	53.8 (273)
TAA Canonical	133	1.6 (93)	0.0 (2)	7.9 (38)
ATE	500	by training frequency of the gold technique: low 62.3, middle 64.7, high 70.5 (tertiles)		

TABLE VII

TASK SCORES AND AGGREGATES FOR THE CANONICAL QWEN2.5-32B CHAIN (0–100; ONE EVALUATION PER CHECKPOINT). TAA IS SHOWN AS CLASSIC/CANONICAL.

Checkpoint	CKT	ATE	RCM	RMS	VSP	TAA C/Can	C	Y	MMLU-Pro	A_{bal}	B_{bal}
Base	52.9	18.2	59.5	7.6	78.8	48.0/2.0	42.6	90.2	69.0	40.3	57.7
Core	72.9	67.2	69.8	63.8	83.1	46.5/18.4	31.1	89.6	59.2	64.9	61.9
+TAA	75.4	63.4	69.0	62.9	83.5	47.0/36.8	29.8	89.4	58.2	66.0	61.7
CSE	72.5	63.4	67.7	63.7	78.9	53.5/3.4	42.0	88.8	54.2	62.4	64.4
Recal-32B	75.5	65.8	67.1	62.5	77.8	50.5/3.4	42.6	89.8	57.4	62.6	65.0

TABLE VIII

PAIRED CHECKPOINT DIFFERENCES (LATER MINUS EARLIER) WITH 95% ITEM-BOOTSTRAP INTERVALS. ASTERISKS INDICATE INTERVALS EXCLUDING ZERO.

Metric	+TAA − Core	CSE − +TAA	Recal-32B − CSE	Recal-32B − Core
CKT	+2.6 [1.0,4.2]*	-3.0 [-4.5,-1.4]*	+3.1 [1.5,4.6]*	+2.7 [1.1,4.2]*
ATE	-3.8 [-7.6,0.2]	0.0 [-4.8,4.6]	+2.4 [-2.4,7.2]	-1.4 [-5.2,2.8]
RCM	-0.8 [-1.3,-0.2]*	-1.3 [-2.2,-0.4]*	-0.6 [-1.9,0.8]	-2.7 [-3.9,-1.5]*
RMS	-1.0 [-3.3,1.6]	+0.9 [-1.5,3.3]	-1.2 [-3.9,1.5]	-1.3 [-4.0,1.4]
VSP	+0.3 [-0.1,0.7]	-4.5 [-5.5,-3.5]*	-1.1 [-2.3,0.0]	-5.3 [-6.4,-4.2]*
TAA Canon.	+18.4 [12.4,24.4]*	-33.5 [-39.8,-27.1]*	0.0 [-3.0,2.6]	-15.0 [-20.7,-9.4]*
CSE malware	+0.6 [-0.4,1.6]	+19.7 [15.3,24.2]*	-1.8 [-4.5,1.0]	+18.6 [14.9,22.5]*
CSE threat intel	-4.5 [-6.7,-2.3]*	+7.9 [3.3,12.4]*	+4.4 [1.8,7.2]*	+7.9 [3.6,12.0]*
MMLU-Pro	-1.0 [-1.7,-0.4]*	-4.0 [-4.8,-3.2]*	+3.2 [2.5,3.9]*	-1.8 [-2.6,-1.1]*
A_{bal}	+1.1 [0.1,2.2]*	-3.6 [-4.8,-2.3]*	+0.2 [-0.9,1.3]	-2.3 [-3.4,-1.1]*
B_{bal}	-0.1 [-0.6,0.4]	+2.7 [1.7,3.8]*	+0.6 [-0.2,1.4]	+3.2 [2.2,4.2]*

from MMLU-Pro. Layering is therefore a trade rather than a uniform improvement. Because the later layers also add the attribution and SOC-format shards, this comparison does not separate the value of that data from the value of training on it in separate layers; a single pass over all four shards with matched tokens and a chain that omits the attribution layer would isolate the two, and were not run for this paper.

D. Why Canonical Attribution Collapses

Aggregate scores cannot distinguish lost knowledge from a formatting change, so every Canonical-TAA response of the retained checkpoints was classified (Table IX). The collapse is semantic: at CSE, every response is well formed and 124 of 133 name a real ATT&CK group that is nevertheless wrong, and G-codes disappear entirely. Recalibration restores G-code formatting but not correctness. The SOC-oriented layer therefore redirects the association behavior itself, an effect visible only because intermediate checkpoints were retained.

E. Where Severity Prediction Regresses

Because VSP is scored by base-score distance, a checkpoint can score 77.8 while matching all eight CVSS base metrics exactly in only 13.6% of items. Table X decomposes the errors. Exact-vector agreement falls from 25.6% at +TAA to 16.4% at CSE and 13.6% at Recal-32B, with errors concentrated in the fields describing how a vulnerability is reached: Attack

TABLE IX
CANONICAL-TAA RESPONSE TAXONOMY (133 ITEMS).

Outcome	+TAA	CSE	Recal-32B
Strict correct	17	4	3
Correct alias	64	1	3
Wrong known group	32	124	113
Other/missing	20	4	14
Contains G-code	132	0	121

TABLE X
VSP ERROR PROFILE OVER 2,000 ITEMS. COLUMNS AFTER "EXACT" ARE COUNTS OF ITEMS MISMATCHING EACH CVSS V3.1 BASE METRIC.

Checkpoint	Exact	AV	AC	PR	UI	S	C	I	A
Core	483	149	127	369	344	163	634	653	341
+TAA	513	136	131	357	342	115	618	661	340
CSE	329	537	125	799	422	421	578	504	356
Recal-32B	271	583	211	791	372	525	601	507	444

Vector mismatches rise from 136 to 537 and 583, Privileges Required from 357 to 799 and 791, and Scope from 115 to 421 and 525, while impact metrics are comparatively stable. These are the fields that most change remediation priority, and the scalar score partly hides the shift.

F. Transfer to Other Base Models

Table XI applies the same recipe to four other base models; we call each such transfer a *port*. The dense Qwen2.5 models gain on every CTI aggregate while losing 11.6–13.9 MMLU-Pro points. The sparse Qwen3 model gains 1.6 points on B_{bal} but loses 24.6 MMLU-Pro points. Its base was scored with thinking mode and its fine-tuned checkpoint without; served with thinking, as its base was, the checkpoint still scores 53.0 on MMLU-Pro, so that loss is not a serving artifact, but the CTI comparison was not repeated under matched serving. Both 8B models decline on B_{bal} despite some CTI gains, so the recipe should not be applied to much smaller or already security-specialized bases without architecture-specific tuning.

G. Hosted Reference Models

Table XII places the final 32B checkpoint against hosted models evaluated with the same harness. The local model is 7.9 points below GPT-5.5 and 1.0 below DeepSeek-V3.2-Exp on B_{bal}, exceeds Gemini-2.5-Flash by 1.6, and on the ten-test CTI mean lies between GPT-5.2 and DeepSeek-V3.2-Exp; its MMLU-Pro score is 20–30 points lower. Evaluating it on all four suites took about $1.42 of local GPU time (two H100s at an assumed $2.50 per GPU-hour), against $508 of API charges for GPT-5.5. This excludes the planned 408–512 GPU-hours of training and is not a lifecycle cost comparison. These are single runs and a reference frame, not a controlled ranking; they indicate narrow CTI specialization rather than broad capability parity.

VIII. DISCUSSION

A. Implications for Cybersecurity SLM Design

Graph-grounded generation and layered IFT address different failure surfaces. Ariadne and Sophia act on supervision: they make the origin of each target inspectable and let coverage be designed around explicit CTI schemas. Retained checkpoints act on optimization: they reveal how later specialization changes behavior earlier layers had acquired. The graph path is directly auditable; against the single-stage first layer, the four-layer order trades aligned accuracy for SOC-format competence (Section VII-C).

For SLMs, limited capacity and aggressive specialization create a tension between narrow competence and general capability. The 32B chain gains strongly on schema-aligned tasks while losing MMLU-Pro, and the 8B transfers show that a recipe that works on a larger base can fail on a smaller one. An SLM is therefore better treated as a bounded specialist whose accepted task types are defined by measured competence, with broader reasoning routed elsewhere.

The transitions also argue for keeping intermediate checkpoints: +TAA is materially better at canonical attribution than later checkpoints, while CSE improves SOC-style tasks. A single sequence can thus move the model between capability profiles rather than improving one objective monotonically. Retained checkpoints let a deployment select the checkpoint that matches its role.

B. Limitations and Operational Boundaries

Scope. The tasks evaluated are the individual mapping steps that feed a risk assessment and mitigation plan. We do not evaluate a complete patch plan produced for a real organization's assets and scan data, and no private data was used.

Benchmark alignment. Training deliberately targets AthenaBench's task formats, and one author co-developed that benchmark. The overlap audit does not establish leakage as the explanation for the gains, but public-catalog overlap, shared answer spaces, paraphrase, and benchmark-guided recipe development limit out-of-sample claims. The final checkpoint does not improve CyberSOCEval and lowers MMLU-Pro. An entity-disjoint or temporally held-out incident set, evaluated once after checkpoint selection on separate development data, is needed for a strong generalization claim.

Causal attribution. Graph-derived data, shard composition, training length, order, sequence length, packing, and learning rate co-vary across layers, so the retained checkpoints show observed transitions, not causal effects. The Core checkpoint provides a single-stage baseline (Section VII-C), but a one-pass mixture over all four shards with matched tokens and a Core→CSE chain that omits attribution were not run, and the label "final" denotes the released checkpoint, not an established optimum.

Uncertainty and reproducibility. Intervals cover item sampling, not training-seed variance: every checkpoint is one training run, and seeded replication of Layer 4 and of the

TABLE XI
CROSS-ARCHITECTURE PORTS UNDER THE UNIFORM AGGREGATION. VALUES COMPARE EACH BASE TO THE PROJECT'S SELECTED CHECKPOINT; ALL ROWS ARE SINGLE TRAINING/EVALUATION RUNS.

		B_{bal}			Ten-test mean			MMLU-Pro		
Base model	Selected checkpoint	Base	Port	Δ	Base	Port	Δ	Base	Port	Δ
Qwen2.5-32B-Instruct	Recal-32B	57.7	65.0	+7.3	49.6	62.9	+13.3	69.0	57.4	-11.6
Qwen2.5-14B-Instruct	Recalibrate	55.1	61.0	+5.9	46.1	59.6	+13.5	64.1	50.2	-13.9
Qwen3-30B-A3B-Thinking	CSE	61.8	63.4	+1.6	54.9	60.9	+6.0	77.6	53.0	-24.6
Foundation-Sec-8B-Instruct	Recalibrate	56.4	53.5	-2.9	52.7	53.2	+0.4	43.2	25.6	-17.6
Llama-3.1-8B-Instruct	Recalibrate	50.7	49.8	-0.9	43.4	50.8	+7.3	48.3	27.7	-20.6

TABLE XII
HOSTED REFERENCES AND THE LOCAL 32B MODELS UNDER THE UNIFORM AGGREGATION (SINGLE EVALUATIONS). C IS THE CYBERSOCEVAL COMPOSITE AND Y THE CYBERMETRIC MEAN.

Model	A_{bal}	C	Y	B_{bal}	Ten-test mean	MMLU-Pro
GPT-5.5	77.0	48.3	93.3	72.9	73.5	88.3
DeepSeek-V4-Pro	67.2	52.2	91.8	70.4	67.9	79.1
Gemini-3-Flash-Preview	69.3	48.1	92.1	69.8	68.4	89.2
GPT-5.2	63.8	46.8	91.9	67.5	65.0	81.3
DeepSeek-V3.2-Exp	59.0	47.2	91.7	66.0	62.2	82.5
Local Recal-32B	62.6	42.6	89.8	65.0	62.9	57.4
Gemini-2.5-Flash	53.1	46.2	91.1	63.4	57.9	82.1
Qwen2.5-32B-Instruct base	40.3	42.6	90.2	57.7	49.6	69.0

full chain would test recipe robustness. Row-level regeneration was not fully seed-pinned (graph-read order and deduplication tie-breaks can vary), so exact replay requires preserving the graph snapshot, serialized shards, and seeds together.

Operational safety. Graph-grounded training does not guarantee graph-faithful answers at inference time; retrieval, constrained decoding, output verification, and human approval are separate mechanisms. The Canonical-TAA collapse and the VSP reachability errors show that the model should not be the sole basis for attribution or prioritization decisions. Prompt-injection resistance, private-telemetry performance, latency under load, and analyst utility were not established.

IX. CONCLUSION AND FUTURE WORK

This paper asked whether a locally served SLM can be specialized for the catalog mappings of vulnerability response without giving up the traceability that security decisions require, and without hiding what specialization costs.

Our answer has four parts. First, Ariadne and Sophia show that IFT data for CTI can be generated without a language model writing the targets: Ariadne merges twelve public sources into a typed graph of 214,651 nodes and 330,935 relationships, and Sophia compiles expert-written templates into graph queries whose results are rendered into triples, producing a 219,675-row core corpus in which every target traces to a template, a graph binding, and public identifiers. Second, layered IFT on Qwen2.5-32B-Instruct turns this data into large gains on the mappings that matter most operationally: attack-technique extraction rises from 18.2 to 65.8 and mitigation selection from 7.6 to 62.5, and the three-benchmark mean reaches 65.0, within a point of DeepSeek-V3.2-Exp and above Gemini-2.5-Flash, from a model served on a two-GPU host. Third, retaining every checkpoint turns interference from an invisible side effect into a measured quantity: the attribution layer adds 18.4 points of canonical attribution that the SOC-oriented layer then erases through wrong but well-formed group names, and the same layer shifts CVSS errors into Attack Vector, Privileges Required, and Scope. Fourth, the overlap audit, extended after training to CyberSOCEval and the general instruction data, finds little verbatim overlap and does not establish leakage as the explanation for the mapping gains.

Three practical lessons follow. Provenance can be designed into training data rather than reconstructed afterwards: when the model learns a wrong mapping, the template and query that taught it can be inspected and corrected. A specialized SLM is best deployed as a bounded specialist for validated task types, with its MMLU-Pro loss treated as a signal to route broader reasoning elsewhere. And a fine-tuning schedule should be evaluated layer by layer, because the checkpoint that is best for one analyst role—here, +TAA for attribution—need not be the final one.

Future work follows directly from these results: one-pass training on all four shards with matched tokens, a chain that omits attribution, and seeded replications, to separate the effect of layering from that of the added data; temporally held-out evaluation, in which a frozen Ariadne snapshot trains a model that is tested on CVEs, techniques, and campaigns that appear only in later snapshots, paired with embedding-based screening for paraphrase; replay mixtures that recover canonical attribution and the CVSS reachability fields without surrendering the CyberSOCEval gain; organization-specific training data generated inside the security boundary from assets, scanner findings, and tickets; verifiable rewards computed by re-executing Sophia queries; and deployment studies

that route task families to the checkpoint best suited to them and measure analyst utility, latency, and prompt-injection resistance.

More broadly, a specialized security model is only as trustworthy as the path from its training data back to the source. Making that path explicit, and measuring what each step of specialization gains and loses, is what allows such a model to be deployed with bounded and documented risk.

ACKNOWLEDGMENT

This work was funded by Athena Software Group Inc., which develops Glaukopis. P. J. Worth Jr. is the company's founder, president, chief executive officer, and board chairman and holds shares in it; I. Cardei advises the company as its Chief Scientist and holds incentive options in it; and P. J. Worth Jr. is a co-author of AthenaBench. OpenAI ChatGPT and Anthropic Claude Code assisted with editing, reference checks, and analysis scripts; the authors verified all content.

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